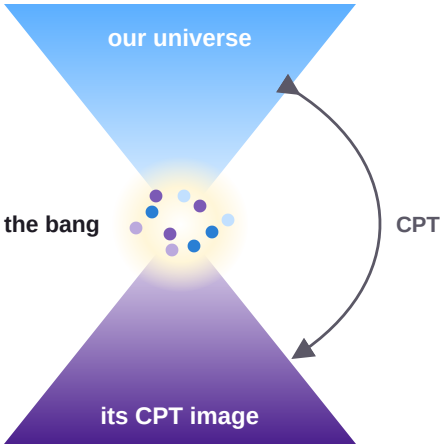
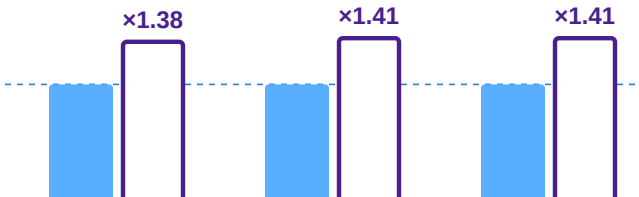
Widely used graviton-exchange rates count the plasma's fermions twice

Separate Ways and the Upside Down: corrected rates, and what they mean for the dark matter of a CPT-symmetric universe



Boyle, Finn and Turok: the universe before the bang is the CPT image of the one after, and its dark matter is a right-handed neutrino, near 480 PeV, made at the bang.

×1.41 too large for fermion dark matter, and **×1.38** for a scalar: the Standard Model's 45 Weyl fermions counted as 45 Dirac fermions.



corrected, in Table 1 published count Section 2, Fig. 1, Appendix C

Without inflation: 491.6 PeV at most

No starting state that respects CPT makes fewer dark-matter particles than Boyle, Finn and Turok's, so their mass is a ceiling, with no floor. Section 4

Hot plasma: 8.0×10^{13} GeV is enough

Even with the corrected rate, in Einstein gravity a plasma that ever reached that temperature makes all the dark matter. Section 5

If it decays: a line at 245.8 PeV at most

A dark-matter neutrino decays to a neutrino line at half its mass. The lightest neutrino is massless, so the masses sum to 58.8 meV. Section 6

Every number is made by public code and checked against the text: doi.org/10.5281/zenodo.23034297